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Thermodynamic assessment of the Al–Cu–Gd system

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ABSTRACT

Phase equilibria of the Cu–Gd and Al–Cu–Gd systems were analyzed, and a complete thermodynamic description of the system was obtained. The thermodynamic descriptions of the Al–Cu and Al–Gd systems were taken from the literature. Most of the binary intermetallic phases except Al_2Gd, Cu_2Gd, and Cu_5Gd were treated as being without solubility in the ternary system. Based on the experimental data in the literature, seven stable ternary intermetallic phases were involved in the modeling of this system. Among them, one ternary compound was treated as a semi-stoichiometric compound with large homogeneity ranges of both Al and Cu. The rest were treated as stoichiometric compounds. The calculated phase equilibria were in agreement with the available experimental data.

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1. Introduction

Interest in aluminum alloys is increasing continuously, especially for their potential applications in the automotive and aerospace industries. Microalloying has been used to improve the high temperature strength of the selected age hardenable Al alloys (such as Al–Cu–Mg–Zn and Al–Cu–Mg–Ag alloys) throughout the years [1–3]. Some rare earth elements in conventional aluminum alloys have shown beneficial effects in the casting process [4–6]. The addition of these elements contributes to a reduction of the content of gases and impurities, and decreases the arm spaces between secondary dendrites. It has been reported that the mechanical properties, such as the incubation time and the peak hardness of Al alloys, did not change after partial replacement of expensive Sc by Gd [7]. And Lantsman et al. [8] reported that Gd could considerably slow down the recrystallization process of Al–Cu-Mn alloys. Moreover, utilization of Gd as an alloying element in Al-based amorphous alloys has been studied for years [9–11]. Not only can the rare earth element Gd benefit the mechanical properties of amorphous alloys [12,13], but also it can improve the glass forming ability (GFA) [14,15]. It has been reported that the mechanical properties, such as the ductility of amorphous alloys, can be improved by introducing Gd into Al–Cu alloys [16].

Knowledge of the thermodynamics and phase diagram is vital in the field of materials research and process control, which is especially important for the development of new alloys. The aims of the present work are then to evaluate Al–Cu–Gd ternary system and develop a self-consistent thermodynamic description by means of the CALculation of PHAse Diagram (CALPHAD) technique [17].

2. Experiment information

2.1. The Al–Cu and Al–Gd binary systems

The Al–Cu system was thermodynamically studied by Saunders [18]. Recently, Witusiewicz et al. [19] reassessed the liquid and $\gamma D8_3$ phases of the Al–Cu phase diagram on the basis of new experimental thermodynamic data concerning the $\gamma D8_3$ phase. In this study, the thermodynamic parameters by Witusiewicz et al. [19] are adopted as is shown in Fig. 1.

The Al–Gd system was first thermodynamically assessed by Grobner et al. [20]. Then, Hackenberg et al. [21] reassessed the Al-rich region of this system when they calculated the Al–Fe–Gd phase diagram. Recently, Cacciamani et al. [22] reassessed the whole system and the reported that thermodynamic parameters were fitting to higher order systems. Therefore, the parameters taken from Cacciamani et al. [22] are used in this work, as shown in Fig. 2.

2.2. The Cu–Gd binary system

Copeland and Kato [23] investigated the Gd-rich region of the Cu–Gd binary system. The terminal solid solubility of Cu in Hcp_Gd

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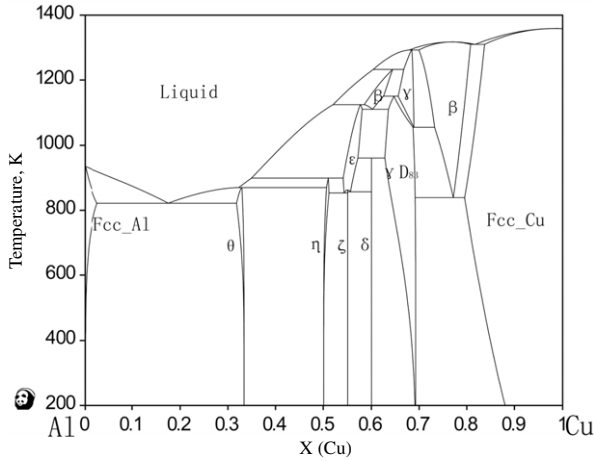

Fig. 1. Calculated Al–Cu phase diagram by Witusiewicz et al. [19].

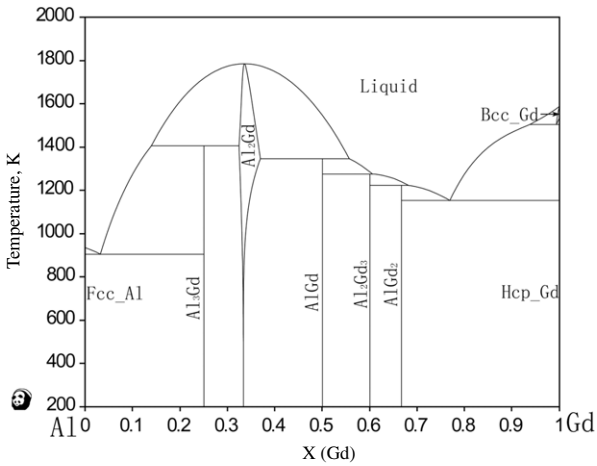

Fig. 2. Calculated Al–Gd phase diagram by Cacciamani et al. [22].

was reported to be less than 0.5 at.%, and a eutectic reaction (Liquid → Hcp_Gd + CuGd) was found as well.

Buschow and Goot [24] reported a high-temperature formation phase Cu_7Gd. According to their result, this phase decomposed to Cu_6Gd and Fcc_Cu below 700 °C.

Subsequently, Cheng and Zeng [25] reported the Cu–Gd phase diagram by means of DTA, XRD and metallography. In their work, four intermetallic compounds were identified: only the Cu_5Gd melts congruently at 932 °C whereas Cu_6Gd, Cu_2Gd and CuGd formed peritectically at 884 °C, 870 °C and 759 °C respectively. Moreover, two eutectic reactions were reported : Liquid → Fcc_Cu + Cu_6Gd (875 °C, 8 at.% Gd); Liquid → CuGd + Gd (668 °C, 68 at.% Gd). Neither the solid solubilities of Gd in Cu nor Cu in Gd were reported in their work.

The Cu–Gd system was systematically determined by Carnasciali et al. [26] through DTA, XRD and metallography. Six compounds were found: Cu_6Gd, Cu_5Gd, Cu_9Gd_2, Cu_7Gd_2, Cu_2Gd and CuGd. Among these phases, Cu_9Gd_2(930 °C), Cu_2Gd(860 °C) and CuGd(830 °C) melted congruently while Cu_6Gd(865 °C), Cu_5Gd(925 °C) and Cu_7Gd_2(870 °C) formed peritectically. A thermal effect at 665 °C corresponding to the stoichiometry of the Cu_5Gd phase was reported which indicated that this phase underwent a structure transformation from the $AuBe_5$-type to $CaCu_5$-type. In addition, the stable temperature range of the Cu_7Gd_2 phase was from 825 °CC to 870 °CC Their investigation also indicated the occurrence of four eutectic reactions : Liquid → Fcc_Cu + Cu_6Gd (860 °C, 9.5 at.% Gd), Liquid → Cu_9Gd_2 + Cu_2Gd (820 °C, 29 at.% Gd), Liquid → Cu_2Gd + CuGd(770 °C, 44 at.%Gd), Liquid → CuGd + Bcc_Gd (675 °C, 70 at.% Gd). In the Gd-rich region, Carnasciali et al. [26] also reported a eutectoid reaction at 630 °C: Bcc_Gd → CuGd + Hcp_Gd. It was found that the solid solubility of Gd in Cu could be ignored while the solubilities of Cu in Bcc_Gd and Hcp_Gd were estimated to be about 15 at.% and 3 at.% Cu, respectively.

Based on the experimental information from Cheng and Zeng [25] and Carnasciali et al. [26], Subramanian and Laughlin [27] assessed the Cu–Gd system. Most of the phase relation information of this binary system from the work of Subramanian and Laughlin [27] is accepted except for the Cu-rich phase Cu_7Gd. Buschow and Goot [24] reported the stoichiometry of the phase in equilibrium with the Fcc-Cu terminal solution was Cu_7Gd. However, Cheng and Zeng [25] examined alloys with compositions between those of Fcc_Cu and Cu_6Gd by thermal analysis and XRD and failed to observe the Cu_7Gd phase. On the other hand, Carnasciali et al. [26] found no evidence of Cu_7Gd phase in their thermal and metallographic. Therefore, the Cu_7Gd phase is treated as an unstable phase in the present work.

The enthalpies of mixing of liquid Cu–Gd alloys measured at 1523 K using high temperature calorimetry were reported by Witusiewicz and Ivanov [28]. Based on the high temperature direct synthesis calorimetry method, Fitzner and Kleppa determined the enthalpies of mixing of Cu–Gd alloys at 1473 K [29], and they also reported the standard enthalpies of formation of the Cu_9Gd_2, Cu_2Gd and CuGd phases. Chatain et al. [30] measured the thermodynamic activity of Cu in the Cu–Gd liquid alloy at 1500 K. Moverover, Sommer et al. [31] determined the heats of formation of the intermetallic compounds of the Cu–Gd system by solution calorimetry in liquid tin at 1100 K. The enthalpies of formation of the Cu_9Gd_2, Cu_2Gd and CuGd phases at 298 K were also calculated by Sommer et al. [31].

2.3. The Al–Cu–Gd ternary system

2.3.1. Ternary compounds in the Al–Cu–Gd system

As summarized by Riani et al. [32], Ran [33] and Raghavan [34], there are seven stable ternary intermetallic compounds that have been reported in the literature. Their structures are listed in Table 1.

Among those, the compound Al_8Cu_4Gd (denoted as τ_1) with the $Mn_{12}Th$ type structure was first reported by Buschow et al. [35], and confirmed later by Felner and Nowik [36]. This phase was also reported by Prevarskii and Kuz'ma [37] and Gumeniuk et al. [38] when they investigated the phase relations of the Al–Cu–Gd system.

The $Al_8Cu_{11}Gd$ phase was indicated by Prevarskii and Kuz'ma [37] with the unknown structure. Later, Gumeniuk et al. [38] proposed that this compound has the composition $Al_{3.2}Cu_{7.8}Gd$ (τ_2) with the $BaCd_{11}$ type structure in the ternary system.

Prevarskii and Kuz'ma [37] found a compound Al_4Cu_5Gd with an unknown structure in this ternary system. Later, Gumeniuk et al. [38] reinvestigated this phase and found it had a $(Al_{0.42}Cu_{0.58})_{11}Tb$ type structure with the approximate composition, $Al_{4.4}Cu_{6.6}Gd$ (τ_3).

The $Al_{10}Cu_7Gd_2$ compound with the Th_2Zn_{17} type structure was first identified by Prevarskiy and Kuz'ma [39]. However, a large homogeneity range of Al in this phase was reported to be 45%–52% [37] and 40%–52% [38] respectively when they investigated the phase diagram of Al–Cu–Gd system. In this work, this phase is denoted as $(Al, Cu)_{17}Gd_2$ (τ_4).

Prevarskii and Kuz'ma [37] reported a compound Al_3CuGd with a Al_4Ba type structure. However, in the work of Gumeniuk et al. [38] the structure of this phase was identified as $Al_{11}La_3$ and the composition was modified to be $Al_{8.9}Cu_{2.1}Gd_3$. In this work, this phase is modeled as $Al_{8.9}Cu_{2.1}Gd_3$ (τ_5).

Table 1
Crystallographic data of intermetallic phases.

Phase	Lattice parameters			Structure type, Pearson symbol, space group
	a (nm)	b (nm)	c (nm)	
θ	0.6067		0.4877	Al_2Cu, *tI* 12, *I*4/*mcm*
η	1.2066	0.4105	0.6913	AlCu, *mC*20, *C*2/*m*
ζ	0.40972	0.71313	0.99793	$Al_9Cu_{11.5}$, *oI* 24, *Imm*2
ε	0.4146		0.5063	In Ni_2, *hP*6, *P*63/*mmc*
γD83	0.87023			Al_4Cu_9, *cP*52, $P\bar{4}3m$
γ				Cu_5Zn_8, *cI* 52, $I\bar{4}3m$
β	0.2946			W, *cI*2, $Im\bar{3}m$
Al_3Gd	0.6331		0.4600	Ni_3Sn, *hp*8, *P*63/*mmc*
Al_2Gd	0.7906			Cu_2Mg, *cF*24, $Fd\bar{3}m$
AlGd	0.5893	1.159	0.5695	AlEr, *oP*16, *Pmma*
Al_2Gd_3	0.8320		0.7628	Al_2Zr_3, *tP*20, *P*42/*mnm*
$AlGd_2$	0.6742	0.5254	0.9756	Co_2Si, *oP*12, *Pnma*
Cu_6Gd	0.8030	0.5020	0.9995	$CeCu_6$, *op*28, *Pnma*
Cu_5Gd_L	0.7060			$AuBe_5$, *cF*24, $F\bar{4}3m$
Cu_5Gd_H	0.5020		0.4120	$CaCu_5$, *hP*6, *P*6/*mmm*
Cu_9Gd_2	0.5000		1.3900	
Cu_7Gd_2				
Cu_2Gd	0.4320	0.6860	0.7330	$CeCu_2$, *oI* 12, *Imma*
CuGd	0.3501			ClCs, *cP*2, $Pm\bar{3}m$
τ_1-Al_8Cu_4Gd	0.8748		0.5146	$Mn_{12}Th$, *tI* 26, *I*4/*mmm*
τ_2-$Al_{3.2}Cu_{7.8}Gd$	1.0269		0.66054	$BaCd_{11}$, *tI* 48, *I*41/*amd*
τ_3-$Al_{4.4}Cu_{6.6}Gd$	1.4303	1.4962	0.6572	$(Al_{0.42}Cu_{0.58})_{11}Tb$, *oF**, *Fddd*
τ_4-$(Al, Cu)_{17}Gd_2$	0.8830		1.2830	Th_2Zn_{17}, *hR*57, $R\bar{3}m$
τ_5-$Al_{8.9}Cu_{2.1}Gd_3$	0.42398	1.2553	0.99393	$Al_{11}La_3$, *oI* 12, *Immm*
τ_6-$Al_{2.1}Cu_{0.9}Gd$	0.55153		2.5519	Ni_3Pu, *hR*36, $R\bar{3}m$
τ_7-AlCuGd	0.7077		0.40649	AlNiZr, *hP*9, *P*62*m*

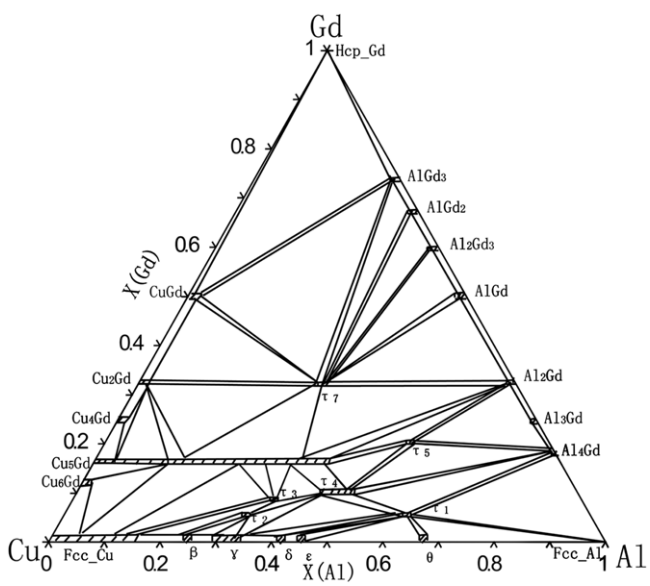

Fig. 3. Measured 773 K isothermal section of Al–Cu–Gd system [37].

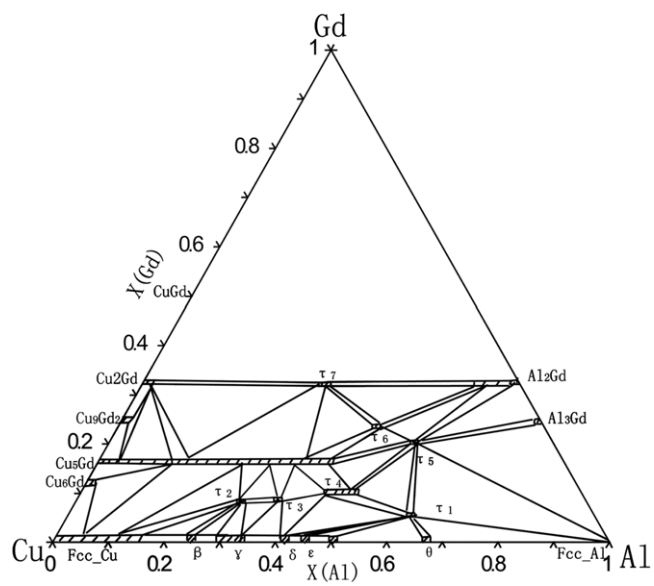

Fig. 4. Measured 873 K isothermal section of Al–Cu–Gd system [38].

The $Al_{2.1}Cu_{0.9}Gd$ compound with the Ni_3Pu type structure was discovered by Gumeniuk et al. [38], which is treated as τ_6 in the present work.

Dwight et al. [40] found an AlCuGd compound (τ_7) with an AlNiZr type structure. Later, it was confirmed by Prevarskii and Kuz'ma [36] and Gumeniuk et al. [38].

2.3.2. Phase equilibria in the solid state

The Al–Cu–Gd ternary isothermal section at 500 °C was reported by Prevarskii and Kuz'ma [37], as shown in Fig. 3. There were six ternary compounds shown in Fig. 4. Among them, Al_8Cu_4Gd, $Al_8Cu_{11}Gd_2$, Al_4Cu_5Gd, Al_3CuGd and AlCuGd are stoichiometric compounds whereas Al and Cu can partially substitute in $(Al, Cu)_{17}Gd_2$ and we call these semi-stoichiometric phases. At 500 °C, the homogeneity range of Al in $(Al, Cu)_{17}Gd_2$ was reported to be 45–52 at.% Al. It was also reported that the homogeneity ranges of Al in Cu_2Gd and Cu_5Gd were 2 at.% Al and 45 at.% Al respectively [37].

Based on the work of Prevarskii and Kuz'ma [37], Gumeniuk et al. [38] reinvestigated this ternary system. In their work, the result of phase analysis of the samples annealed at 500 °C and 600 °C showed the same Al–Cu–Gd ternary compounds at both temperatures. And their experimental result is shown in Fig. 4. From Fig. 4, Gumeniuk et al. [38] proposed seven ternary compounds in this system, one more than that of Prevarskii and Kuz'ma [37], and the structure of every ternary compound was determined with X-ray powder diffraction patterns. Similar to the result of Prevarskii and Kuz'ma [37], except for $(Al, Cu)_{17}Gd_2$, other ternary compounds are stoichiometric compounds. The homogeneity range of Al at 600 °C in $(Al, Cu)_{17}Gd_2$ was determined

to be 40–52 at.% Al. The solubility of Al in Cu_2Gd is about 3 at.% Al while Cu in Al_2Gd is about 5 at.% Cu. Also it is reported that the solubility of Al in the Cu_5Gd_H phase reaches 45 at.% Al.

So far, there is no available thermodynamic description of the Al–Cu–Gd system in the literature. Therefore, the Al–Cu–Gd system has been assessed on the basis of the above mentioned ternary phase diagram information.

3. Thermodynamic model

3.1. Solution phases

The lattice stabilities for elemental Al, Cu and Gd are referred to by Dinsdale [41]. An ordinary substitutional solution model is employed to describe Liquid, Fcc, Bcc and Hcp terminal solutions. The mole Gibbs energy of a solution phase $\Phi(\Phi = \text{Liquid, Fcc, Bcc, Hcp})$ can be represented as the sum of the weighted Gibbs energy for the pure components with the ideal entropy term describing a random mixing of the components and the excess Gibbs energy describing the degree of deviation from ideal mixing, i.e.

$$G^{\phi} = \sum_{i=\text{Al,Cu,Gd}} x_i^0 G_i^{\phi} + RT \sum_{i=\text{Al,Cu,Gd}} x_i \ln(x_i) + {}^{ex}G^{\phi}. \quad (1)$$

Where

$$\begin{aligned} {}^{ex}G^{\phi} &= x_{\text{Al}}x_{\text{Cu}} \sum_{j=0,1,\ldots}^{N} (x_{\text{Al}} - x_{\text{Cu}})^{j\,(j)}L^{\phi}_{\text{Al,Cu}} \\ &+ x_{\text{Al}}x_{\text{Gd}} \sum_{j=0,1,\ldots}^{N} (x_{\text{Al}} - x_{\text{Gd}})^{j\,(j)}L^{\phi}_{\text{Al,Gd}} \\ &+ x_{\text{Cu}}x_{\text{Gd}} \sum_{j=0,1,\ldots}^{N} (x_{\text{Cu}} - x_{\text{Gd}})^{j\,(j)}L^{\phi}_{\text{Cu,Gd}} + x_{\text{Al}}x_{\text{Cu}}x_{\text{Gd}}L^{\phi}_{\text{Al,Cu,Gd}} \end{aligned} \quad (2)$$

where ϕ denotes the solution phases, x_i (i = Al, Cu and Gd) denote the mole fraction of component i, and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structural state of ϕ.

${}^{(j)}L^{\phi}_{\text{Al,Cu}}$ is taken from Witusiewicz et al. [19] and ${}^{(j)}L^{\phi}_{\text{Al,Gd}}$ is taken from Cacciamani et al. [22]. ${}^{(j)}L^{\phi}_{\text{Cu,Gd}}$ is expressed as:

$${}^{(j)}L^{\phi}_{\text{Cu,Gd}} = A_j + B_jT \quad (3)$$

where the A_j and B_j are constants to be optimized.

$L^{\phi}_{\text{Al,Cu,Gd}}$ is ternary interaction parameters. Due to lack of experiment data, it is set to zero.

3.2. Binary intermetallic phases

Most of the binary intermetallic phases are treated as stoichiometric phases, which are described as M_pN_q. According to the Neumann–Kopp rule, the Gibbs energy of the phase M_pN_q is formulated as:

$$G_{M_pN_q} = \frac{p}{p+q}{}^0G_M^{HSER} + \frac{q}{p+q}{}^0G_N^{HSER} + A + BT. \quad (4)$$

Where M and N denote Al, Cu, or Gd, A and B are the adjusted parameters to be optimized in the present work.

According to Refs. [37,38], the Al_2Gd, Cu_5Gd_H and Cu_2Gd phases exhibit the homogeneity regions in the Al–Cu–Gd ternary system. The Gibbs energy expression of Cu_5Gd and Cu_2Gd phases are formalized as

$$\begin{aligned} G^{(Al,Cu)_xGd_y} &= Y^I_{\text{Al}}G_{\text{Al:Gd}} + Y^I_{\text{Cu}}G_{\text{Cu:Gd}} \\ &+ \frac{x}{x+y}RT(Y^I_{\text{Al}}\ln Y^I_{\text{Al}} + Y^I_{\text{Cu}}\ln Y^I_{\text{Cu}}) \\ &+ Y^I_{\text{Al}}Y^I_{\text{Cu}}L_{\text{Al,Cu:Gd}}. \end{aligned} \quad (5)$$

Where x and y are the stoichiometry ratios. The superscript I denotes the first sublattice. Y^I_{Al} and Y^I_{Cu} stand for the site fractions of Al, Cu in the first sublattices, respectively. And the term $L_{\text{Al,Cu:Gd}}$ represents the interaction between the Al and Cu in the first sublattice, expressed by the Redlich–Kister polynomials. The parameter $G_{\text{Cu:Gd}}$ is the Gibbs energy of formation of the compound Cu_xGd_y, which can be taken from the binary system. And $G_{\text{Al:Gd}}$ is expressed as

$$G_{\text{Al:Gd}} = \frac{x}{x+y}{}^0G^{\text{Fcc}}_{\text{Al}} + \frac{y}{x+y}{}^0G^{\text{Hcp}}_{\text{Gd}} + A + BT. \quad (6)$$

Which represents the Gibbs energy of the assumed Al_xGd_y compound with Cu_xGd_y structure.

A and B are the adjustable parameters to be optimized in the present work.

The Gibbs energy expression of Al_2Gd phase is formalized as

$$\begin{aligned} &G^{(\text{Al,Cu,Gd})_2(\text{Al,Cu,Gd})_1} \\ &= \sum_i\sum_j Y^I_iY^{II}_jG_{i:j} + 2RT(Y^I_{\text{Al}}\ln Y^I_{\text{Al}} + Y^I_{\text{Cu}}\ln Y^I_{\text{Cu}} + Y^I_{\text{Gd}}\ln Y^I_{\text{Gd}}) \\ &+ RT(Y^{II}_{\text{Al}}\ln Y^{II}_{\text{Al}} + Y^{II}_{\text{Cu}}\ln Y^{II}_{\text{Cu}} + Y^{II}_{\text{Gd}}\ln Y^{II}_{\text{Gd}}) \\ &+ \sum_i\sum_j\sum_k Y^I_iY^I_jY^{II}_k \sum_{v=0,1,\ldots} {}^vL_{i,j:k}(Y^I_i - Y^I_j)^v \\ &+ \sum_i\sum_j\sum_k Y^I_kY^{II}_iY^{II}_j \sum_{v=0,1,\ldots} {}^vL_{k:i,j}(Y^{II}_i - Y^{II}_j)^v. \end{aligned} \quad (7)$$

Where i, j, k denote Al, Cu, Gd, and $L_{i,j:k}$ stands for the interaction between the i and j in the first sublattice while the second sublattice is fully occupied by the element k. $G_{\text{Al:Al}}$, $G_{\text{Al:Gd}}$, $G_{\text{Gd:Al}}$ and $G_{\text{Gd:Gd}}$ are taken from the corresponding binary systems. Other parameters are to be optimized in this work.

3.3. Ternary intermetallic compounds

In the ternary system, τ_1-Al_8Cu_4Gd, τ_2-$Al_{3.2}Cu_{7.8}Gd$, τ_3-$Al_{4.4}Cu_{6.6}Gd$, τ_5-$Al_{8.9}Cu_{2.1}Gd_3$, τ_6-$Al_{2.1}Cu_{0.9}Gd$ and τ_7-AlCuGd are modeled as stoichiometric phases ($Al_xCu_yGd_z$), i.e. Al, Cu, and Gd cannot substitute each other in any sublattice. τ_2– $(\text{Al, Cu})_{17}Gd_2$ are treated as semi-stoichiometric phases ($(\text{Al, Cu})_xGd_y$), i.e. Al and Cu can partially substitute each other in the first sublattice, but Al and Cu cannot substitute Gd atoms in the second sublattice. The Gibbs energy expression for each of these stoichiometric compounds is written as

$$\begin{aligned} G_{\text{Al}_x\text{Cu}_y\text{Gd}_z} &= \frac{x}{x+y+z}{}^0G^{\text{Fcc}}_{\text{Al}} + \frac{y}{x+y+z}{}^0G^{\text{Fcc}}_{\text{Cu}} \\ &+ \frac{z}{x+y+z}{}^0G^{\text{Hcp}}_{\text{Gd}} + A + BT. \end{aligned} \quad (8)$$

Where x, y and z are the stoichiometry ratios of the sublattices. And for the semi-stoichometric compounds, it becomes

$$\begin{aligned} G^{(Al,Cu)_xGdy} &= Y^I_{\text{Al}}G_{\text{Al:Gd}} + Y^I_{\text{Cu}}G_{\text{Cu:Gd}} \\ &+ \frac{x}{x+y}RT(Y^I_{\text{Al}}\ln Y^I_{\text{Al}} + Y^I_{\text{Cu}}\ln Y^I_{\text{Cu}}) \\ &+ Y^I_{\text{Al}}Y^I_{\text{Cu}}L_{\text{Al,Cu:Gd}}. \end{aligned} \quad (9)$$

Where

$$G_{\text{Al:Gd}} = \frac{x}{x+y}{}^0G^{\text{Fcc}}_{\text{Al}} + \frac{y}{x+y}{}^0G^{\text{Hcp}}_{\text{Gd}} + A + BT \quad (10)$$

$$G_{\text{Cu:Gd}} = \frac{x}{x+y}{}^0G^{\text{Fcc}}_{\text{Cu}} + \frac{y}{x+y}{}^0G^{\text{Hcp}}_{\text{Gd}} + A + BT. \quad (11)$$

Where the $L_{\text{Al,Cu:Gd}}$ represents the interaction between Al and Cu in the first sublattice, expressed by the Redlich–Kister polynomials. A and B are the adjusted parameters to be optimized in the present work.

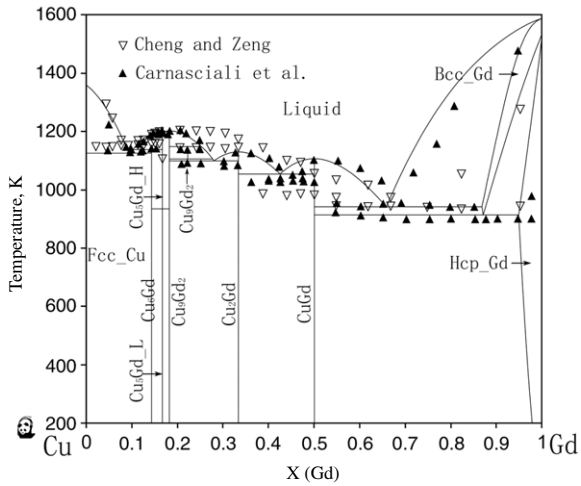

Fig. 5. Calculated phase diagram of Cu–Gd binary system with experimental data [25,26].

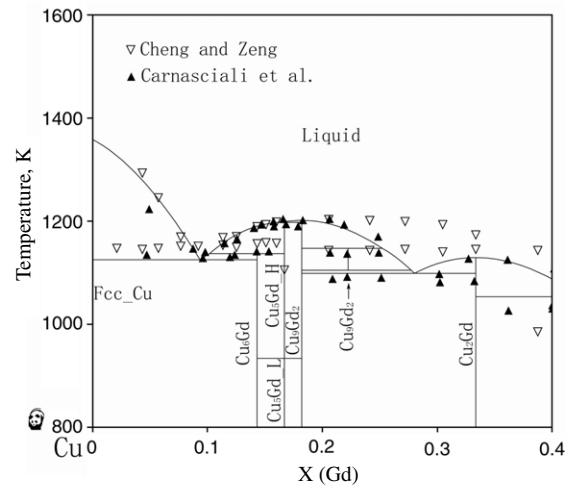

Fig. 6. Calculated Cu-rich phase diagram with experimental data [25,26].

4. Results and discussion

4.1. The Cu–Gd binary system

On the basis of lattice stabilities cited from Dinsdale [41], the optimization of the Cu–Gd system is carried out using the PanOptimizer of Pandat software [42]. PanOptimizer is a software package for evaluating the thermodynamic, kinetic and thermophysical model parameters from experimental or theoretically calculated data by solving the weighted least squares problem. More detailed information about PanOptimizer can be found in Ref. [43]. The phase diagram and thermochemical data were used as input to the program for the optimization. Each piece of selected information was given a certain weight by personal judgment, and verified by trial and error method during the assessment, until most of the selected experimental information was reproduced. All the evaluated parameters are listed in Table 2.

Figs. 5 and 6 show the calculated phase diagram of Cu–Gd system compared with experimental data. Through the comparison, it is clear that our calculated results are in good agreement with the work of Carnasciali et al. [26]. Table 3 lists the invariant reactions of the Cu–Gd system. An agreement within 9 K between calculated and experimentally determined temperatures for these invariant reactions has been obtained. The assessed maximum terminal solubility of Cu in α Gd and β Gd is 4 and 14 at.% respectively, which agrees well with the experimental data (3 and 15 at.% Cu) [26].

As illustrated in Fig. 7, the measured enthalpies of mixing of liquid Cu–Gd alloys [28,29] are reproduced by using the optimized parameters. The minimum values of enthalpies of mixing in the Cu–RE system (such as Cu–Dy [44], Cu–Er [45] and Cu–Yb system [46]) are around 40 at.% RE, and our work fits this tendency well. In addition, the thermodynamic activity of Cu in the Cu–Gd liquid alloy at 1500 K is calculated, as shown in Fig. 8, which is in good agreement with the experimental data [30]. It is found that the experimental data of thermodynamically properties can be described well by the present thermodynamic parameters within experimental error.

Calculated standard enthalpies of formation of intermetallic phases at 298 K are compared with experimental data in Fig. 9. As can be seen, calculated results are reasonably consistent with the experimental data measured by Fitzner and Kleppa [29], but a discrepancy exists when compared with the data of Sommer et al. [31] Actually, in our assessment, we found that it was very difficult to fit the result of Sommer et al. [31] with other experimental data. This problem also exists in the assessment of

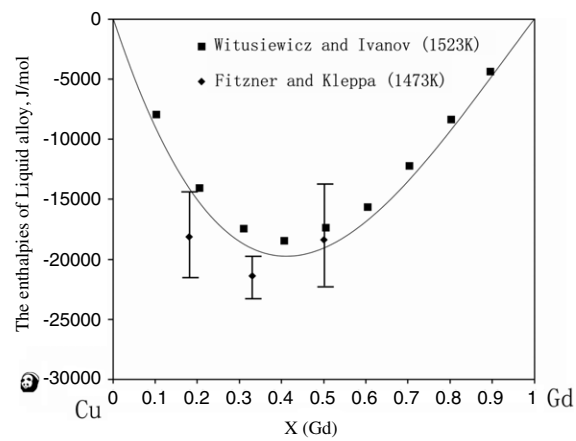

Fig. 7. Calculated mixing enthalpies of of Cu–Gd liquid alloys at 1523 K with experimental data [28,29] (referred to liquid state).

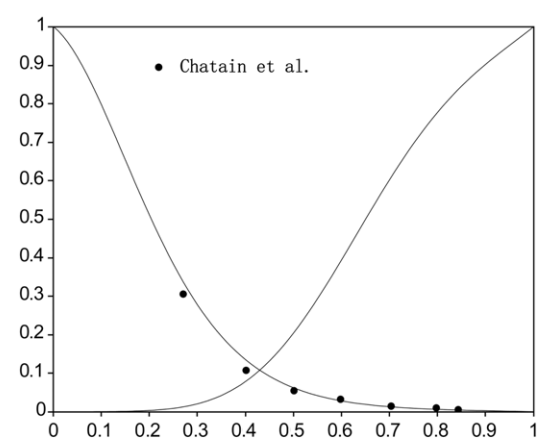

Fig. 8. Calculated thermodynamic activities of Cu–Gd liquid alloys at 1500 K with experimental data [30] (referred to liquid state).

Cu–Dy [44] and Cu–Er [45] systems. Fitzner and Kleppa's data was reported in 1997, whereas the work of Sommer et al. [31] was published in 1986. So in this work, the experimental data measured by Fitzner and Kleppa was adopted during the assessment, and the data by Sommer et al. [31] was just used for comparison.

Table 2
Thermodynamic parameters of Al–Cu–Gd system.

Phase	Thermodynamic parameters	Ref
Liquid model: (Al, Cu, Gd)	${}^{0}L^{Liq}_{Al,Cu} = -67094 + 8.555T$	[19]
	${}^{1}L^{Liq}_{Al,Cu} = +32148 - 7.118T$	[19]
	${}^{2}L^{Liq}_{Al,Cu} = +5915 - 5.889T$	[19]
	${}^{3}L^{Liq}_{Al,Cu} = -8175 + 6.049T$	[19]
	${}^{0}L^{Liq}_{Al,Gd} = -166500 + 52.36T$	[22]
	${}^{1}L^{Liq}_{Al,Gd} = -23790$	[22]
	${}^{2}L^{Liq}_{Al,Gd} = 18520$	[22]
	${}^{0}L^{Liq}_{Cu,Gd} = -76316.8 + 1.0721T$	This work
	${}^{1}L^{Liq}_{Cu,Gd} = -30000$	This work
Bcc Model: (Al, Cu, Gd)	${}^{0}L^{Bcc}_{Al,Cu} = -73554 + 4T$	[18]
	${}^{1}L^{Bcc}_{Al,Cu} = +51500 - 11.84T$	[18]
	${}^{0}L^{Bcc}_{Al,Gd} = -80000 + 30.64T$	[22]
	${}^{0}L^{Bcc}_{Cu,Gd} = -31184.5 + 0.005T$	This work
	${}^{1}L^{Bcc}_{Cu,Gd} = +14950.9$	This work
Hcp Model: (Al, Cu, Gd)	${}^{0}L^{Hcp}_{Cu,Gd} = 11742.3 + 6.7981T$	This work
	${}^{0}L^{Hcp}_{Cu,Gd} = 50612.3$	This work
Al_2Gd Model: $(Al, Cu, Gd)_2$ $(Al, Cu, Gd)_1$	$G^{Al2Gd}_{Al:Al} = 15000 + 3GHSERAL$	[22]
	$G^{Al2Gd}_{Gd:Gd} = 15000 + 3GHSERGD$	[22]
	$G^{Al2Gd}_{Cu:Cu} = 15000 + 3GHSERCU$	This work
	$G^{Al2Gd}_{Al:Gd} = -160000 + 32.64T + 2GHSERAL + GHSERGD$	[22]
	$G^{Al2Gd}_{Gd:Al} = +196000 - 32.64T + GHSERAL + 2GHSERGD$	[22]
	$G^{Al2Gd}_{Al:Cu} = +30000 + 2GHSERAL + GHSERCU$	This work
	$G^{Al2Gd}_{Cu:Al} = +30000 + GHSERAL + 2GHSERCU$	This work
	$G^{Al2Gd}_{Cu:Gd} = -56000 + 4.24T + 2GHSERCU + GHSERDY$	This work
	$G^{Al2Gd}_{Gd:Cu} = +86000 - 4.24T + GHSERCU + 2GHSERDY$	This work
	${}^{0}L^{Al2Dy}_{Al,Cu:Gd} = -37000$	This work
Cu_6Gd Model: $(Cu)_{0.857143}(Gd)_{0.142857}$	$G^{Cu6Gd}_{Cu:Gd} = -13652.5 + 0.857143GHSERCU + 0.142857GHSERGD$	This work
Cu_5Gd_LModel : $(Cu)_{0.833333}(Gd)_{0.166667}$	$G^{Cu5Gd_L}_{Cu:Gd} = -15862.17 + 0.011T + 0.833333GHSERCU + 0.166667GHSERGD$	This work
Cu_5Gd_HModel : $(Al, Cu)_{0.833333}(Gd)_{0.166667}$	$G^{Cu5Gd_H}_{Cu:Gd} = -15852.17 + 0.833333GHSERCU + 0.166667GHSERGD$	This work
	$G^{Cu5Gd_H}_{Al:Gd} = -13881.6 + 0.833333GHSERAL + 0.166667GHSERGD$	This work
	$G^{Cu5Gd_H}_{Al,Cu:Gd} = -69881.6$	This work
Cu_9Gd_2 Model : $(Cu)_{0.818182}(Gd)_{0.181818}$	$G^{Cu9Gd2}_{Cu:Gd} = -17294.9 + 0.818182GHSERCU + 0.181818GHSERGD$	This work
Cu_7Gd_2 Model : $(Cu)_{0.777778}(Gd)_{0.222222}$	$G^{Cu7Gd2}_{Cu:Gd} = -11064.7 - 6.602T + 0.777778GHSERCU + 0.222222GHSERGD$	This work
Cu_2Gd Model : $(Al, Cu)_{0.6667}(Gd)_{0.3333}$	$G^{Cu2Gd}_{Al:Gd} = -10257.6 + 1.5T + 0.6667GHSERAL + 0.3333GHSERGD$	This work
	$G^{Cu2Gd}_{Cu:Gd} = -19991.5 - 1.9T + 0.6667GHSERCU + 0.3333GHSERGD$	This work
	${}^{0}L^{Cu2Gd}_{Al,Cu:Gd} = -41457$	This work
CuGd Model: $(Al, Cu)_{0.5}(Gd)_{0.5}$	$G^{CuGd}_{Cu:Gd} = -17985.9964 - 3.5578T + 0.5GHSERCU + 0.5GHSERGD$	This work
τ_1 Model: $(Al)_{0.615385}(Cu)_{0.307692}$ $(Gd)_{0.076923}$	$G^{\tau 1}_{Al:Cu:Gd} = -31500 + 1.4T + 0.615385GHSERAL + 0.307692GHSERCU + 0.076923GHSERGD$	This work
τ_2 Model: $(Al)_{0.266667}(Cu)_{0.65}$ $(Gd)_{0.083333}$	$G^{\tau 2}_{Al:Cu:Gd} = -28500 + 1.3T + 0.266667GHSERAl + 0.65GHSERCU + 0.083333GHSERGD$	This work
τ_3 Model: $(Al)_{0.366667}(Cu)_{0.55}$ $(Gd)_{0.083333}$	$G^{\tau 3}_{Al:Cu:Gd} = -31000 + 1.3T + 0.366667GHSERAL + 0.55GHSERCU + 0.083333GHSERGD$	This work
τ_4 Model: $(Al, Cu)_{0.894737}(Gd)_{0.105263}$	$G^{\tau 4}_{Al;Gd} = -12050 + 1.7T + 0.894737GHSERAL + 0.105263GHSERGD$	This work
	$G^{\tau 4}_{Cu:Gd} = -5950 + 1.9T + 0.894737GHSERCU + 0.105263GHSERGD$	This work
	$G^{\tau 4}_{Al,Cu:Gd} = -82900$	This work
τ_5 Model: $(Al)_{0.635714}(Cu)_{0.15}$ $(Gd)_{0.214286}$	$G^{\tau 5}_{Al:Cu:Gd} = -40500 + 1.3T + 0.0.635714GHSERAL + 0.15GHSERCU + 0.214286GHSERGD$	This work
τ_6 Model: $(Al)0.525(Cu)_{0.225}$ $(Gd)_{0.25}$	$G^{\tau 6}_{Al:Cu:Gd} = -42300 + 1.3T + 0.525GHSERAL + 0.225GHSERCU + 0.25GHSERGD$	This work
τ_7 Model: $(Al)_{0.333333}(Cu)0.333333(Gd)_{0.333334}$	$G^{\tau 7}_{Al:Cu:Gd} = -42300 + 1.3T + 0.333333GHSERAL + 0.333333GHSERCU + 0.333334GHSERGD$	This work

4.2. The Al–Cu–Gd ternary system

Because there is no available thermochemical and liquidus projection information, the ternary interaction parameters of liquid and terminal solid solutions are considered as zero. Based on the phase relations reported by Gumeniuk et al. [38], the Al–Cu–Gd ternary system is assessed. The optimized parameters are listed in Table 2.

The calculated isothermal sections at 773 K and 873 K for the Al–Cu–Gd system are shown in Figs. 10 and 11, respectively.

Table 3
Invariant reactions in the Cu–Gd system.

Reaction	Composition of liquid (at.% Gd)	T/K	Type	Ref
L ⇔ Fcc_Cu + Cu_6Gd	9.3	1125	Eutectic	This work
	9.5	1133	Eutectic	[26]
	9.0	1148	Eutectic	[25]
L + Cu_5Gd_H ⇔ Cu_6Gd	10.3	1139	Peritectic	This work
	10.0	1138	Peritectic	[26]
	9.5	1159	Peritectic	[25]
L + Cu_9Gd_2 ⇔ Cu_5Gd_H	16.2	1196	Peritectic	This work
	16.7	1198	Peritectic	[26]
Cu_5Gd_H ⇔ Cu_5Gd_L	–	935	Allotropic	This work
	–	933	Allotropic	[26]
L ⇔ Cu_9Gd_2	–	1201	Congruent	This work
	–	1203	Congruent	[26]
	–	1205	Congruent	[25]
L + Cu_9Gd_2 ⇔ Cu_7Gd_2	25.2	1147	Peritectic	This work
	26.5	1143	Peritectic	[26]
Cu_7Gd_2 ⇔ L + Cu_9Gd_2	27.7	1105	Catatectic	This work
		1098	Catatectic	[26]
L ⇔ Cu_9Gd_2 + Cu_2Gd	28.0	1098	Eutectic	This work
	29.0	1093	Eutectic	[26]
	60.5	1143	Peritectic	[25]
L ⇔ Cu_2Gd	–	1129	Congruent	This work
	–	1133	Congruent	[26]
L ⇔ Cu_2Gd + CuGd	42.5	1052	Eutectic	This work
	44.0	1043	Eutectic	[26]
		1032	Peritectic	[25]
L ⇔ CuGd	–	1106	Congruent	This work
	–	1103	Congruent	[26]
L ⇔ CuGd + Bcc_Gd	65.8	941	Eutectic	This work
	70	948	Eutectic	[26]
	68.5	941	Eutectic	[25]
Bcc_Gd ⇔ CuGd + Hcp_Gd		912	Eutectoid	This work
		903	Eutectoid	[26]

Comparing our calculated results with the experimental data [37, 38], it can be found that the intermetallic phases presented in the calculated three boundary systems are not the same as shown in the Refs. [37,38].

According to the calculated Al–Cu binary phase diagram, as shown in Fig. 1, β and ε phases are metal-stable at 773 K, while they are stable in the isothermal section reported by Prevarskii and Kuz'ma [37]. Similarly, θ phase existing at 873 K according to Gumeniuk et al. [38] is inconsistent with the binary Al–Cu system where a liquid phase exists there. The similar problems also exist in the Al–Gd boundary system, Al_4Gd and $AlGd_3$ phase were suggested by Prevarskii and Kuz'ma [37] as stable phases which disagreed with the calculated binary Al–Gd phase diagram [22], where these two phases are unstable in the whole temperature ranges. Later, when Ran [33] and Raghavan [34] assessed this ternary system, the isothermal sections were corrected in order to eliminate the above inconsistencies, β and ε were not taken into consideration at 773 K, and the liquid phase was presented instead of θ phase, Al_4Gd and $AlGd_3$ were eliminated at two temperatures. From the work of Ran [33] and Raghavan [34], phase relationships along the Al–Cu and Al–Gd boundary systems would be in better agreement with the accepted binary systems. Therefore, the assessed results of Ran [33] and Raghavan [34] are accepted in the present work.

Moreover, the Cu_9Gd_2 phase was observed as Cu_4Gd phase in the work of Prevarskii and Kuz'ma [37]. In our assessment, it is treated as Cu_9Gd_2 phase. Also the Cu_5Gd phase was determined to be of the $CaCu_5$-type (denoted as Cu_5Gd_H in this work) at both 773 K and 873 K [37,38], however, according to the binary Cu–Gd system, this phase should be the $AuBe_5$-type (denoted as Cu_5Gd_L in this work) at these temperatures. In order to be consistent with the binary system, Cu_5Gd with the $AuBe_5$-type structure was treated as Cu_5Gd_L at 773 K and 873 K.

For the same reason, the homogeneity range for Al in the Cu_5Gd_H at 773 K and 873 K is proposed to be 0.05–41 at.% Al (Figs. 10–12) instead of 0–45 at.% reported by the Refs. [37,38].

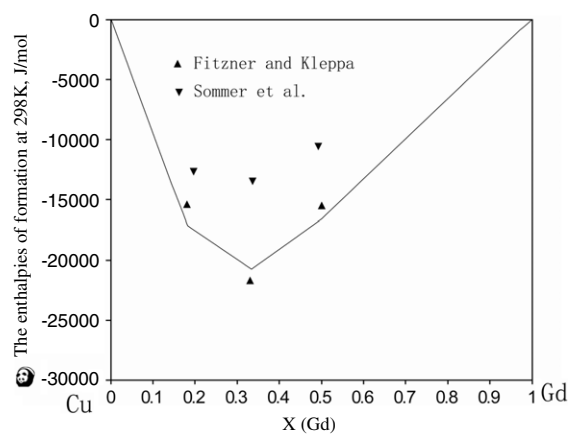

Fig. 9. Calculated enthalpies of formation at 298 K with experimental data [29,31] (reference: Fcc_Cu, Hcp_Gd) (298 K).

The homogeneity region of Al_2Gd and Cu_2Gd extend, respectively, up to 6.5 at.% Cu and 2.5 at.% Al, which are in good agreement with the Refs. [37,38]. In Refs. [37,38], τ_4-$(Al, Cu)_{17}Gd_2$ were determined to contain about 45–52 at.% Al at 773 K and 40–52 at.% Al at 873 K,respectively. From our calculation, the homogeneity range of this phase is 44–49 at.% Al and 41–51 at.% Al, which is in agreement with experimental results.

Moreover, the liquidus projection is calculated and shown in Figs. 13 and 14. Note here the invariant equilibria involving the Al–Cu binary phases are not visible in Fig. 13 because they are very close to the Al–Cu binary subsystem. Further experimental data is needed to verify the liquidus projection.

5. Conclusions

The Al–Cu–Gd ternary system has been assessed thermodynamically based on the reported experimental data of the phase

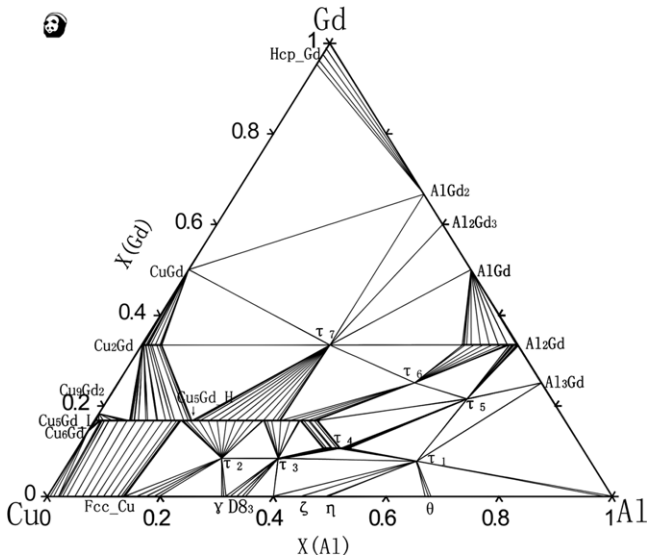

Fig. 10. Calculated isothermal section of Al–Cu–Gd system at 773 K.

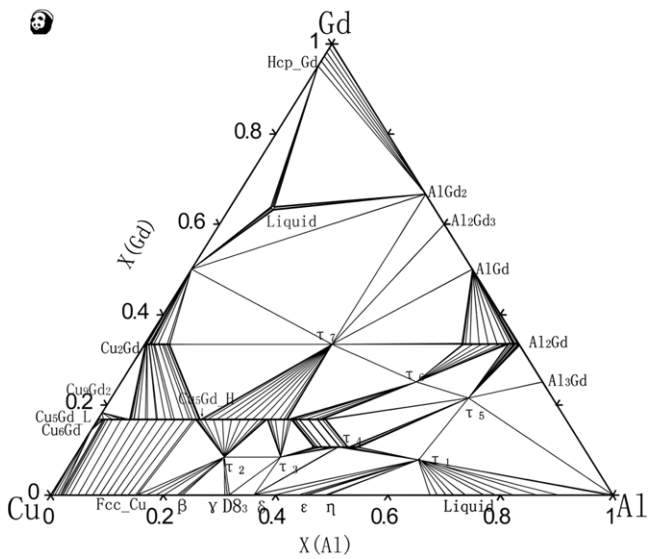

Fig. 11. Calculated isothermal section of Al–Cu–Gd system at 873 K.

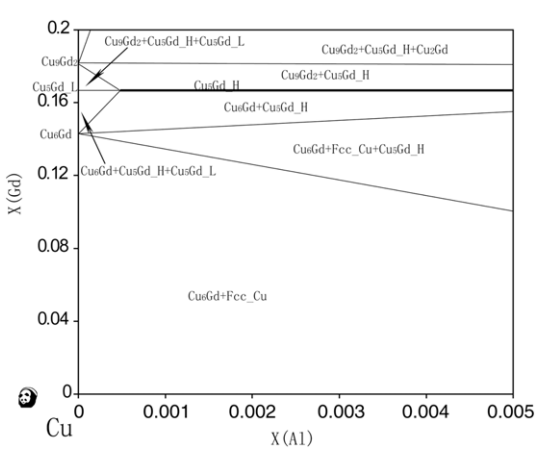

Fig. 12. Calculated isothermal section of Cu-rich region at 873 K.

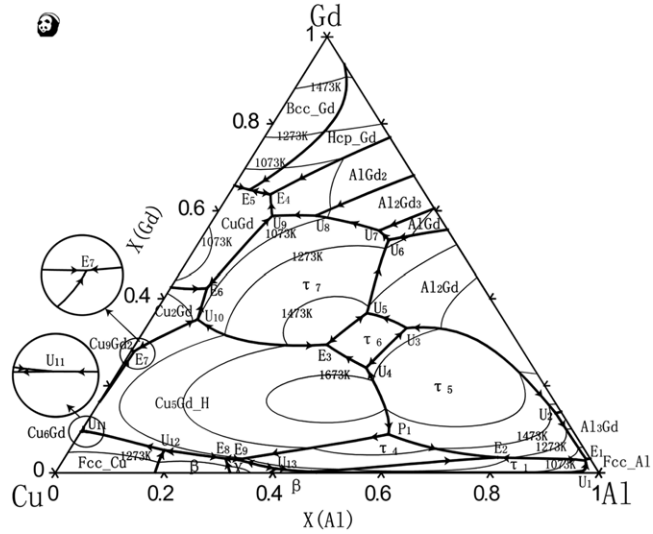

Fig. 13. Calculated liquidus projection of Al–Cu–Gd system with isothermal temperature line.

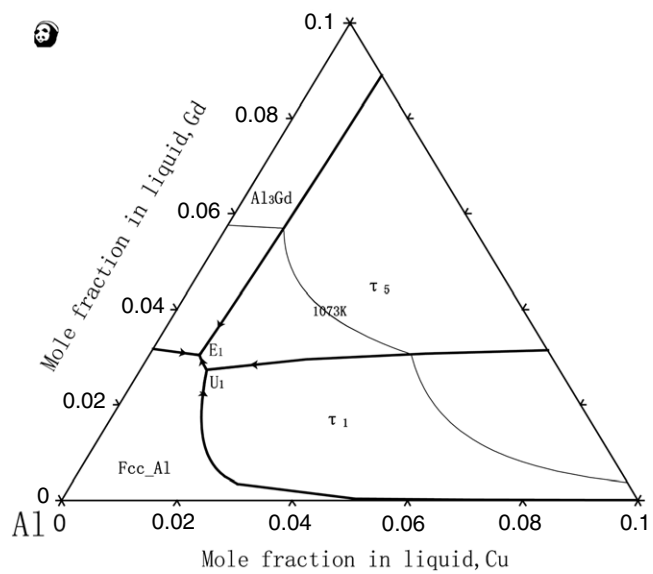

Fig. 14. Calculated liquidus projection of Al-rich corner.

diagrams and the thermodynamic properties. Reasonable agreement between calculation and experimental data has been achieved and the thermodynamic parameters for the various phases in this ternary system have been obtained.

Acknowledgment

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Appendix. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.calphad.2009.08.003.

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